PYTHON LIST – SHORT & COMPLETE NOTES:

1. LIST

A **List** is a collection of multiple values stored in a single variable.

Syntax

a = [10, 20, 30]

Example

a = [10, 20, 30]
print(a)

Output

[10, 20, 30]

---

2. LIST FEATURES

- **Multiple values** → `[10, 20, 30]`
- **Different data types** → `[10, "Ajith", 5.5, True]`
- **Duplicates allowed** → `[10, 20, 10]`
- **Ordered** → maintains insertion order
- **Mutable** → values can be changed

a = [10, 20, 30]
a[1] = 50

print(a)

Output

[10, 50, 30]

---

3. INDEXING

Python indexing starts from **0**.

```
      10   20   30
Index   0    1    2
       -3   -2   -1
```

Positive Index

```
a = [10, 20, 30]

print(a[0])
print(a[1])
```

Output

```
10
20
```

Negative Index

```
print(a[-1])
```

Output

```
30
```

---

4. LIST METHODS

List methods are functions used with a list.

append()
extend()
insert()
remove()
pop()
clear()
copy()
index()
count()
reverse()
sort()

---

5. append()

Adds **one element** at the end.

Syntax

a.append(value)

Example

a = [1, 2, 3]
a.append(4)

print(a)

Output

[1, 2, 3, 4]

Important

a.append([4, 5])

Output:

[1, 2, 3, [4, 5]]

append() → one element

---

6. extend()

Adds **multiple elements** at the end.

a = [1, 2, 3]
a.extend([4, 5])

print(a)

Output

[1, 2, 3, 4, 5]

extend() → multiple elements

---

7. insert()

Adds an element at a **specific index**.

Syntax

a.insert(index, value)

Example

a = [10, 20, 30]
a.insert(1, 100)

print(a)

Output

[10, 100, 20, 30]

---

8. remove()

Removes an element using its **value**.

a = [10, 20, 30]
a.remove(20)

print(a)

Output

[10, 30]

remove() → value

If duplicates exist, it removes the **first occurrence**.

---

9. pop()

Removes an element using its **index**.

```
a = [10, 20, 30]
a.pop(1)

print(a)
```

Output

[10, 30]

Without index

```
a.pop()
```

Removes the **last element**.

pop(index) → index
pop() → last element

---

10. clear()

Removes **all elements**.

```
a = [10, 20, 30]
a.clear()

print(a)
```

Output

[ ]

---

11. copy()

Creates a **copy of the list**.

```
a = [10, 20, 30]
b = a.copy()

b.append(40)

print(a)
print(b)
```

Output

```
[10, 20, 30]
[10, 20, 30, 40]
```

Original list is not changed.

---

12. index()

Returns the **index of a value**.

a = [10, 20, 30, 40]

print(a.index(30))

Output

2

index() → where is the value?

---

13. count()

Returns how many times a value occurs.

a = [10, 20, 10, 30, 10]

print(a.count(10))

Output

3

count() → how many times?

---

14. reverse()

Reverses the **original list**.

a = [10, 20, 30, 40]

a.reverse()

print(a)

Output

[40, 30, 20, 10]

reverse() → changes original list

---

15. sort()

Arranges the original list in **ascending order**.

a = [30, 10, 40, 20]

a.sort()

print(a)

Output

[10, 20, 30, 40]

Descending order

a.sort(reverse=True)

sort() → changes original list

---

16. BUILT-IN FUNCTIONS

Important built-in functions used with lists:

len()
max()
min()
sum()
any()
all()
sorted()
reversed()

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17. len()

Finds the **number of elements**.

```
a = [10, 20, 30]

print(len(a))
```

Output:

3

---

18. max()

Finds the **largest value**.

```
a = [10, 50, 20]

print(max(a))
```

Output:

50

---

19. min()

Finds the **smallest value**.

a = [10, 50, 20]

print(min(a))

Output:

10

---

20. sum()

Finds the **sum of numeric values**.

a = [10, 20, 30]

print(sum(a))

Output:

60

---

21. any()

Returns `True` if **at least one** element is truthy.

a = [False, False, True]

print(any(a))

Output:

True

any() → at least one True

---

22. all()

Returns `True` if **all** elements are truthy.

a = [True, True, True]

print(all(a))

Output:

True

all() → all True

---

23. sorted()

`sorted()` is a **built-in function**.

It returns a **new sorted list**.

a = [30, 10, 40, 20]

b = sorted(a)

print(a)
print(b)

Output

[30, 10, 40, 20]
[10, 20, 30, 40]

Descending

b = sorted(a, reverse=True)

sorted() → original list unchanged

---

24. reversed()

`reversed()` is a **built-in function**.

It returns elements in reverse order.

a = [10, 20, 30, 40]

b = reversed(a)

print(list(b))

Output

[40, 30, 20, 10]

reversed() → original list unchanged

---

25. MOST IMPORTANT DIFFERENCES

append() vs extend()

append() → one element
extend() → multiple elements

remove() vs pop()

remove() → value
pop() → index

index() vs count()

index() → position of value
count() → number of occurrences

reverse() vs reversed()

reverse() → method
changes original list

reversed() → built-in function
original list unchanged

sort() vs sorted()

sort() → method
changes original list

sorted() → built-in function
original list unchanged

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26. ONE-PAGE REVISION

List Methods

Method	Use
`append()`	Add one element
`extend()`	Add multiple elements
`insert()`	Add at specific index
`remove()`	Remove by value
`pop()`	Remove by index
`clear()`	Remove all
`copy()`	Create copy
`index()`	Find index
`count()`	Count occurrences
`reverse()`	Reverse original
`sort()`	Sort original

Built-in Functions

Function	Use
`len()`	Number of elements
`max()`	Largest value
`min()`	Smallest value
`sum()`	Total
`any()`	At least one True
`all()`	All True
`sorted()`	New sorted list
`reversed()`	Reverse iterator